

Please provide a short paragraph describing your thoughts on the required reading below. This can be in any form: tell me what you liked, didn't like, understood, didn't understand. There is no wrong answer as long as you read and reflect.

- Diaz, K., Robinson, T. L., Aydin, Y. O., Aydin, E., Goldman, D. I., & Wan, K. Y. (2021). A minimal robophysical model of quadriflagellate self-propulsion. *Bioinspiration & Biomimetics*, 16(6), 066001.

I found this paper really interesting because it connects motion at a microscale with robotics in order to see what capabilities could be used in robotics from nature. The author showed very easily that simply changing the coordination (gait) of the flagella, without changing the structure, can significantly affect swimming performance. It was easy to see how the trot gait was consistently the fastest in both real algae and the robot made the results feel that this gait was the most efficient out of all of them. I like how the paper also ties back concepts from lecture like low Reynolds number and the need to break symmetry change to achieve motion. One thing I found confusing was understanding how the phase differences between the flagella translate into differences in net displacement, but the figures helped to see this more clearly. Overall, I like the use of figures and structure of the paper. It was very well written and it was easy to follow. Using an actual robot as the physical model was a really effective approach because it allowed us to test various ideas that would be difficult to isolate in real organisms.